

PSORIASIS DETECTION AND PASI SCORING USING DEEP LEARNING

Presented by

Lubdhak Nairith Saha Arnish
B.tech CSE, 3rd year

PROBLEM STATEMENT

Psoriasis is a chronic skin disease that requires accurate detection and severity estimation. Existing systems mostly perform only classification and do not estimate severity.





OBJECTIVE

- Detect psoriasis using deep learning
- Estimate severity using PASI scoring
- Build a combined system



DATASET

- Dataset contains:
 - Psoriasis images
 - Normal (Unknown_Normal) images
- Total distribution:
 - Psoriasis: 820 images
 - Normal: 1651 images

Problem:

- Dataset imbalance

Solution:

- Balanced sampling (300 images per class)

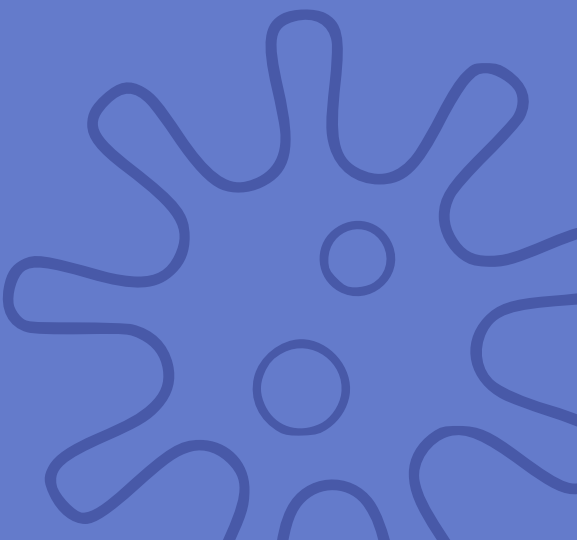




PREPROCESSING

- Resized images to 224×224
- Applied data augmentation:
 - Horizontal flip
 - Rotation
- Applied normalization

Purpose:

- Improve model performance
 - Reduce overfitting
- 
- 

MODEL USED

ResNet18 (Pretrained model)
Transfer Learning applied

Steps:

- **Frozen backbone layers**
- **Trained only final classification layer**

Reason:

- **Limited dataset**
- **Faster and stable training**

TRAINING

- Loss Function: CrossEntropyLoss
- Optimizer: Adam
- Epochs: 3

Result:

- Loss decreased gradually
- Model trained successfully

```
warnings.warn(msg)
Epoch 1, Loss: 0.4900
Epoch 2, Loss: 0.3041
Epoch 3, Loss: 0.2584
Test Accuracy: 80.0%
```

RESULTS

Test Accuracy: 89.0 %
Prediction: Psoriasis

--- Running PASI Scoring ---

Area: 0.362

Erythema: 0.808

Scaling: 0.739

Thickness: 0.469

Final PASI Score: 0.731

PASI SCORING

After detecting psoriasis, severity is calculated using:

Area (size of lesion)

Erythema (redness)

Scaling (texture variation)

Thickness (edge intensity)





PASI CALCULATION PROCESS

Steps:

1. Segment lesion using LAB color space
2. Extract features from lesion region
3. Compute PASI score

Final Output:

- Detection + Severity score

(Add mask + overlay image here)



LIMITATIONS

During the implementation, I initially tried to use more advanced deep learning models such as EfficientNet and segmentation models like U-Net to improve accuracy and lesion detection. However, due to the lack of GPU support and limited computational resources, training these heavy models on CPU was very slow and impractical. Additionally, the dataset size was relatively small and imbalanced, which led to overfitting when using deeper architectures. Because of these limitations, I shifted to a lightweight pretrained ResNet18 model using transfer learning, which provided stable performance and faster training. This allowed me to successfully implement the detection system and further extend it by adding PASI scoring for severity estimation.



FUTURE WORK

In future work, I plan to improve the system by exploring more advanced deep learning models such as EfficientNet for better classification accuracy and U-Net-based architectures for precise lesion segmentation. I also aim to enhance the PASI scoring by using adaptive or region-wise analysis instead of a global approach, which can provide more accurate severity estimation. Additionally, I plan to optimize the model using techniques like pruning or partial fine-tuning to improve efficiency without increasing computational cost. These improvements can make the system more accurate, robust, and suitable for real-world clinical applications.



The background is a solid blue color. It features several stylized virus particles, which are circular with spikes and internal dots, in various sizes and positions. On the left side, there is a semi-circle filled with white diagonal lines. On the right side, there is a large graphic consisting of two semi-circles stacked vertically; the top one is orange and the bottom one is yellow.

THANK YOU